

	particle's momentum and charge. For ions with the same charge but different momentum, its radius will be different.
4(a)(i)	Incident sound wave from the signator is <u>reflected off</u> the metal plate. There are two <u>progressive sound wave of similar frequency and amplitude, moving in opposite direction</u> . They <u>superpose</u> and have <u>destructive and constructive interference, resulting in nodes and anti nodes</u> , forming stationary waves.
4(a)(ii)	period, $T = (5 \text{ cm})(0.50 \text{ ms cm}^{-1}) = 2.5 \text{ ms}$ $f = \frac{1}{T} = \frac{1}{2.5 \times 10^{-3}} = 400 \text{ Hz}$

4(a)(iii)	$v = f\lambda$ $\lambda = \frac{340}{400} = 0.850 \text{ m}$ $\text{distance between adjacent nodes} = \frac{1}{2}\lambda = \frac{1}{2}(0.850) = 0.425 \approx 0.43 \text{ m}$
4(b)(i)	$\text{path difference} = 1.8 - 1.4 = 0.4 \text{ m} = \frac{0.4}{0.2} = 2\lambda$ The sources of the wave are in phase; the <u>path difference between the 2 waves is an integer number of wavelengths</u>, the two waves arriving at P will be <u>in phase</u>. They will superpose, have constructive and destructive interference, resulting in maxima and minima at P.
4(b)(ii)	$I \propto \frac{1}{r^2}$ $\frac{I_x}{I_y} = \left(\frac{r_y}{r_x}\right)^2$ $\frac{4.5 \times 10^{-6}}{I_y} = \left(\frac{1.8}{1.4}\right)^2$ $I_y = 2.7222 \times 10^{-6} \approx 2.7 \times 10^{-6} \text{ W m}^{-2}$
4(b)(iii)	$I \propto A^2$ $\frac{I_x}{I_y} = \left(\frac{A_x}{A_y}\right)^2$ $\frac{4.5 \times 10^{-6}}{2.7222 \times 10^{-6}} = \left(\frac{A_x}{A_y}\right)^2$ $\frac{A_x}{A_y} = 1.2857 \approx 1.3$
5(a)	The resistance of the thermistor is the ratio of the potential difference (p.d.)